THE DEGREE-N PART OF A FINITELY GENERATED GRADED MODULE IS A FINITE MODULE OVER THE DEGREE-ZERO RING

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Allowed standard prerequisites (editorial).

- Homogeneous generators for a finitely generated graded module. Used at the first line of the counted proof (lines 14097-14099). Reduces the generating set to homogeneous elements by taking homogeneous components.
- Graded generation of the algebra over its degree-zero part. Used at proof lines 14097-14100. Produces the homogeneous generators f_i of S over S_0, which can be taken in positive degrees.

Lemma 0.1. *If M is a finitely generated graded S-module, and if S is finitely generated over S_0, then each M_n is a finite S_0-module.*

Proof. Suppose the generators of M are m_i and the generators of S are f_i. By taking homogeneous components we may assume that the m_i and the f_i are homogeneous and we may assume $f_i \in S_+$. In this case it is clear that each M_n is generated over S_0 by the "monomials" $\prod f_i^{e_i} m_j$ whose degree is n. □

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